

AthDGC · Athens-PROIEL

An open diachronic Greek treebank with Indo-European parallels

Nikolaos Lavidas

June 5, 2026

The **Athens Diachronic Glossa Chronos** (*Athens-PROIEL (the dependency-treebank (a collection of sentences whose grammatical structure has been analysed and stored) standard for early Indo-European languages, developed at Oslo)*) is an open PROIEL-style dependency treebank spanning Greek diachrony with Indo-European parallels.

NKUA · Division of Language-Linguistics, Department of English Language and Literature, School of Philosophy

Funded by HFRI (Project No. 20577) · Greece 2.0 NRRP

AthDGC is **both** a continuously-updated **PROIEL XML 2.0** (the file format that stores each sentence as a tree of word-by-word grammatical relations, developed at Oslo) treebank **and** an open-source **computational diachronic-linguistics toolkit**:

- LightSIDE (an open-source text-mining workbench developed at Carnegie Mellon)-compatible feature extraction + classifier training
- NoSketch-style concordancer (a search tool that shows every occurrence of a word in its surrounding context)
- Fine-tuned (further trained on new data to adapt it to a particular text type) Stanza (Stanford's open-source Python pipeline that automatically tags, lemmatises, and parses sentences) checkpoints (Byzantine, Late-Byzantine, Modern adaptations)
- Argument-structure extractor
- Cross-lingual alignment (matching corresponding words or sentences between

Covers:

- Homeric and Archaic Greek (Homer, Hesiod, Sappho, Pindar)
- Classical (Aeschylus, Sophocles, Euripides, Aristophanes, Plato, Aristotle)
- Koine and NT (Plutarch, Strabo, Lucian, the New Testament)
- Late Antique (Eusebius, Basil, Chrysostom, Procopius)
- Byzantine (Psellos, Anna Komnene, John of Damascus, Niketas Choniates)

with **NT verse-level cross-alignment** to Latin (Vulgate), Gothic (Wulfila), and Old Church Slavonic (Marianus).

Metric	Value
Total corpus rows	89.9 M
Annotated Greek rows	4.08 M
NT-aligned Greek verses	6,861
IE cross-aligned witnesses	4

Pipeline

1. **Discovery** - daily harvest from archive.org, Perseus, First1K, Wikisource, Diorisis
2. **Filtering** - Greek-script ratio, apparatus-criticus rejection, dedup
3. **Conversion** - PROIEL XML 2.0
4. **Annotation** - Stanza `grc_proiel` + analogous models per parallel language
5. **Argument-structure** - subject, object, oblique, voice, aspect per verb
6. **Cross-alignment** - LaBSE (a multilingual sentence-embedding model that maps sentences from many languages into a common vector space) + AwesomeAlign (a word-alignment procedure that uses multilingual-BERT attention to match words across translations) (mBERT)
7. **Storage** - PROIEL XML + JSONL (a plain-text data format that stores one record per line) + Qdrant + Neo4j

- **Retranslation** of influential texts (Iliad, NT, Septuagint Psalms, classical historiography)
- **Retelling** chains across periods and languages
- **Argument structure** (the set of obligatory and optional partners a verb requires, e.g. subject, object, oblique) under retranslation

- Lavidas (PI) · Nikiforidou · Haug (Oslo) · Kulikov (Ghent)
- Geka · Symeonidis · Michalareas
- Chionidi · Tsiropina · Plakoutsi · Argyropoulos

NKUA · Athens Digital Glossa Chronos Research Network

Lavidas, N., Nikiforidou, K., Haug, D., Kulikov, L., Geka, V., Symeonidis, V., Michalareas, T., Chionidi, S., Tsiropina, A., Plakoutsi, E., and Argyropoulos, E. (2026). *AthDGC: Athens Diachronic Glossa Chronos*. Zenodo (an open research-data repository hosted at CERN that mints permanent DOIs).

DOI (Digital Object Identifier, a permanent web link for a published artefact):
[10.5281/zenodo.20439182](https://doi.org/10.5281/zenodo.20439182)

- Showcase: <https://athdgc.github.io>
- Samples: <https://athdgc.github.io/samples.html>
- Tools: <https://athdgc.github.io/tools.html>
- Press kit: <https://athdgc.github.io/press.html>
- Launch report: https://athdgc.github.io/data_paper.html
- GitHub: <https://github.com/AthDGC>
- Hugging Face (a public hosting platform for machine-learning models and datasets) mirror: <https://huggingface.co/AthDGC> (in setup; ships at v0.5)

Funded by the **Hellenic Foundation for Research and Innovation (HFRI)** under the 3rd Call for HFRI Research Projects to support Post-Doctoral Researchers, **Project No. 20577**; with complementary support from the **Greece 2.0 National Recovery and Resilience Plan**.

Compute supplied by **GRNET ARIS (the Greek national high-performance computing cluster, run by GRNET)** (Greek national HPC (high-performance computing, i.e. a cluster of fast machines used for heavy computation)), allocation pa260305.